

Sihat Afnan

PhD Student @ University of California, Irvine

Research Areas: Security & Privacy of Embodied AI, Robot Learning and Teleoperation, AR/VR Systems, Autonomous Driving, Adversarial Machine Learning

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Education

University of California, Irvine

Sep 2024 - Present

Ph.D. in Computer Science; GPA: 3.97/4.00

Advisor: [Dr. Habiba Farrukh](#) (SPECT Lab)

Bangladesh University of Engineering and Technology

Apr 2018 - Jun 2023

B.Sc. in Computer Science and Engineering; GPA: 3.81/4.00

Selected Publications

1 Who Moved the Robot? Humanoid Datasets Remember Their Operators

2026

Sihat Afnan, Unnat Jain, Habiba Farrukh

[NeurIPS 2026](#) (Accepted; Atlanta, GA, USA, Dec 2026) | [Project page](#)

Motion retargeting is assumed to be a lossy, *one-way* projection: mapping an operator onto a shared robot skeleton normalizes body proportions and is treated as anonymizing. We show the map is effectively **invertible**. **UNVEIL**, a skeleton-aware spatiotemporal GNN, recovers **operator gender, identity, and age/height** from retargeted joint-angle trajectories alone, generalizing to unseen operators, tasks, and retargeting implementations. Demonstrated on **BONES-SEED** (142K motions, 522 actors, retargeted to the **Unitree G1** by NVIDIA), making shared humanoid motion corpora a **biometric fingerprint**.

2 Who's Watching the Road? Revisiting Physical Adversarial Attacks on Autonomous Driving in the Presence of Human Drivers

2027

Sihat Afnan, Mutahar Ali, Chi Zhang, Praneetsai Vasu Iddamsetty, Mohammad Al Faruque, Qi Alfred Chen, Habiba Farrukh

Under review at [USENIX Security 2027](#)

Revisits **physical adversarial attacks** on autonomous driving with the human driver back in the loop, since prior work evaluates these attacks only against the AV stack itself. Built a high-fidelity **CARLA-based VR simulator** integrating a full AV pipeline (*sensor simulation, ML perception, planning, control*) with **real-time eye tracking** and behavioral logging, and used it to measure whether and how quickly drivers notice and recover from *stop-sign manipulation, lane-detection failure, and phantom obstacles*.

3 Manipulating XR Spatial Understanding Using On-Device Speakers

2027

Sihat Afnan, Habiba Farrukh

Under review at [IEEE S&P 2027](#)

First **on-device acoustic attack** on XR spatial understanding: an optimization framework drives a headset's own built-in speakers to perturb *camera odometry, RGB imaging, and depth sensing*, producing controlled geometric distortions in the reconstructed 3D map with no added hardware. Demonstrated object addition/removal, surface misclassification, and degraded user task performance across **Meta Quest 3S, Apple iPad, and Project Aria glasses**, corrupting over **91%** of spatial maps in real-world Quest 3S trials.

4 LogShield: A Transformer-based Advanced Persistent Threat Detection System Leveraging Self-Attention

2023

*Sihat Afnan**, *Mushtari Sadia**, *Shahrear Iqbal*, *Anindya Iqbal* (*equal contribution)

[Arxiv Pre-print](#)

Self-attention framework for detecting **APT** patterns in event sequences derived from system provenance graphs, reaching **F1 of 98% and 95%** on DARPA OpTC and DARPA TC E3 against **96% and 94%** for LSTM baselines.

Work Experience

Graduate Student Researcher, University of California, Irvine Jun 2025 - Present
| SPECT Lab

Graduate Teaching Assistant, University of California, Irvine Sep 2024 - May 2025
| CS 130: Introduction to Computer Security, CS 122A: Introduction to Data Management, ICS 33: Intermediate Programming with Python, ICS H32: Python Programming with Libraries (Accelerated).

Lecturer, Dept. of CSE, BRAC University – Dhaka, Bangladesh Jun 2023 - Aug 2024
| CSE341: Microprocessors, CSE340: Computer Architecture, CSE321: Operating Systems, CSE230: Discrete Mathematics.

ML Engineer, Spectrum Software & Consulting (Pvt.) Ltd. Feb 2023 - May 2023
| Built ML components for *trade finance* and *invoice financing* products, automating manual document review.

Academic Service

- **Artifact Evaluation Committee Member**, NDSS 2027 (Network and Distributed System Security Symposium).
- **Program Committee Member**, BEA-ACL 2023 Conference Workshop.

Awards & Honors

Computer Science Department Research Fellowship (UCI): Awarded for outstanding research contributions.

International Grammatical Error Detection Competition (Kaggle) – Champion

NLP Hackathon 2023 (Bangladesh Open Source Network) – 2nd Runner Up

HackNSU 2020 (NSU ACM Student Chapter) – 1st Runner Up

University Dean's List Scholarship (BUET): Awarded for academic excellence across Level 1 to Level 4 of undergraduate study.

University Merit List Scholarship (BUET): Awarded for top academic performance in the department.

Technical Skills

Programming Languages: Python, C/C++, C#, Java, Go, Swift, SQL

Deep Learning Frameworks: PyTorch, TensorFlow

XR & Robotics: Unity, Unreal Engine, ARKit, Meta Quest SDK, Project Aria, LiDAR/depth sensing

Simulation & Networking: CARLA, NS-3, Mininet, P4

Systems & Developer Tools: Linux, Git, Django, Arduino, Raspberry Pi, ATmega32

Graduate Coursework at UCI

Robotics Deep Learning (CS295), Computer and Systems Security (CS205), Network and Distributed Systems Security (CS203), Machine Learning (CS273A), Systems and ML (CS256), Information Retrieval (CS221), Image Understanding (CS216), Embedded and Ubiquitous Systems (CS244), Usable Security and Privacy (CS204), Computer Networks (CS232), Research in HCI (INF232).